

INTERNATIONAL BACCALAUREATE

PHYSICS HL

EXTENDED ESSAY

Candidate code: To be filled in

**Investigation on the frequencies of oscillations of a coupled
magnetic-mechanical oscillator**

Word count: 4000

1 Introduction

1.1 Coupled magnetic mechanical oscillator

Simple harmonic motion refers to the motion a body undergoes when the restoring force is directly proportional to the displacement from equilibrium.

A *simple harmonic oscillator* is a system that oscillates with simple harmonic motion. For example, a mass attached to an elastic spring that is freely subjected to gravity is a simple harmonic oscillator.

A *coupled harmonic oscillator* consists of two or more simple harmonic oscillators that are connected together, such that energy can be transferred between them.

In this extended essay, the *coupled magnetic mechanical oscillator* consists of two identical copper leaf springs that are clamped vertically at their lower ends to a rigid and non-magnetic base, as shown in Figure 1. A permanent neodymium magnet is attached to the free end of each spring. The two magnets are oriented so that they repel, hence the like poles face each other.

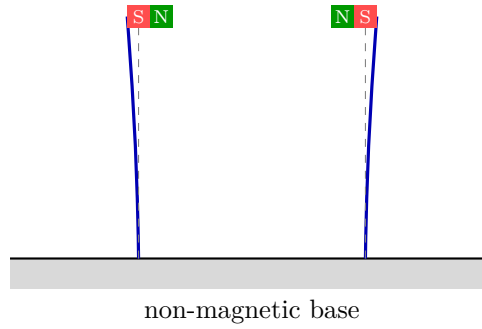


Figure 1: Coupled magnetic mechanical oscillator. Both springs are slightly bent outward by the static repulsion force from the magnets.

1.2 Oscillation modes and frequencies